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Introduction to Robotics Project
Autonomous Mobile Robot using ROS2 and micro-ROS

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Industrial Context

Visual markers such as ArUco markers are widely used in industrial robotics.
Manufacturing: Robots equipped with vision systems detect markers to locate parts or tools on assembly lines. This enables accurate manipulation and placement, improving reliability and repeatability in automated production.
Logistics: Warehouse automation systems use visual markers for localization and navigation. Robots can move between storage locations, identify items, and transport goods efficiently, improving inventory management and order fulfillment.

1. Project Overview

The objective of this project is to design and implement an autonomous mobile robot using a distributed robotics architecture based on ROS2 and micro-ROS.

The system is composed of two computing layers:

- A Raspberry Pi running ROS2 for high-level processing
- An ESP32 microcontroller running micro-ROS for low-level motor control

The robot operates in an indoor environment representing a simplified warehouse. Localization and navigation are achieved using visual detection of ArUco markers placed in the environment. **You may propose alternative localization approaches if desired.**

2. System Architecture

The robot will be divided into two computational layers.

2.1 Low-Level Control (ESP32)

Hardware: ESP32   **Software**: micro-ROS
Responsibilities:

- Motor control using PWM
- Differential drive motor commands
- Communication with ROS2 through micro-ROS

The ESP32 receives velocity commands from ROS2 through the topic `/cmd_vel`.

2.2 High-Level Control (Raspberry Pi)

Hardware: Raspberry Pi 4   **Software**: ROS2
Responsibilities:

- micro-ROS Agent
- Camera acquisition
- ArUco marker detection
- Robot pose estimation
- Navigation between stations
- Object color detection
- Task management
- Visualization using RViz

3. Environment

The robot operates in a structured indoor environment composed of several stations:

- Home position (robot start)
- Manufacturing station
- Storage A
- Storage B

The stations are placed in an open workspace where the robot can move freely between locations.
ArUco markers are placed at known positions in the environment. Each marker has a unique ID and a predefined position in the map.
The robot detects these markers using its onboard camera and uses them as visual landmarks for localization.

Manufacturing Station

The manufacturing station contains small boxes representing produced parts. These boxes can be either **red** or **green**.
The robot must detect the color of the box using its onboard camera and pick it using a mechanical picking mechanism **designed by you**.

Depending on the detected color:

- **Red box** → deliver to **Storage A**
- **Green box** → deliver to **Storage B**

After delivering the box, the robot must return to the **Home position**.

4. Localization

Localization is based on visual detection of ArUco markers.

ArUco Marker Detection

A camera mounted on the robot detects ArUco markers placed in the environment.
For each detected marker, the system estimates:

- Marker ID
- Distance to the marker
- Relative orientation

Since the global position of each marker is known, the robot can estimate its own pose in the environment.
Whenever a marker is detected, the robot updates its estimated position and uses this information for navigation.

5. Functional Objectives

The robot must autonomously perform the following sequence:

1. Start from the Home position
2. Navigate to the Manufacturing station
3. Detect the color of the box
4. Pick the box using the designed mechanism
5. Deliver the box to the correct storage station
6. Return to the Home position

Navigation is based on waypoint following between stations using the pose estimated from ArUco marker detections.

6. Software Structure

The ROS2 system may include the following nodes:

- `base_controller` (micro-ROS node on ESP32)
- `camera_node` / `aruco_detector`
- `color_detector`
- `localization_node`
- `navigation_node`
- `task_manager` (state machine)

Possible data flow:

- Camera → ArUco detection
- ArUco detection → Robot pose estimation
- Camera → Color detection
- Robot pose → Navigation
- Navigation → `/cmd_vel`

8. Deliverable

A fully autonomous robot capable of:

- Navigating between stations
- Detecting box color
- Picking and transporting boxes
- Delivering them to the correct storage station
- Returning to the Home position

Appendix

Warehouse (for demonstration)

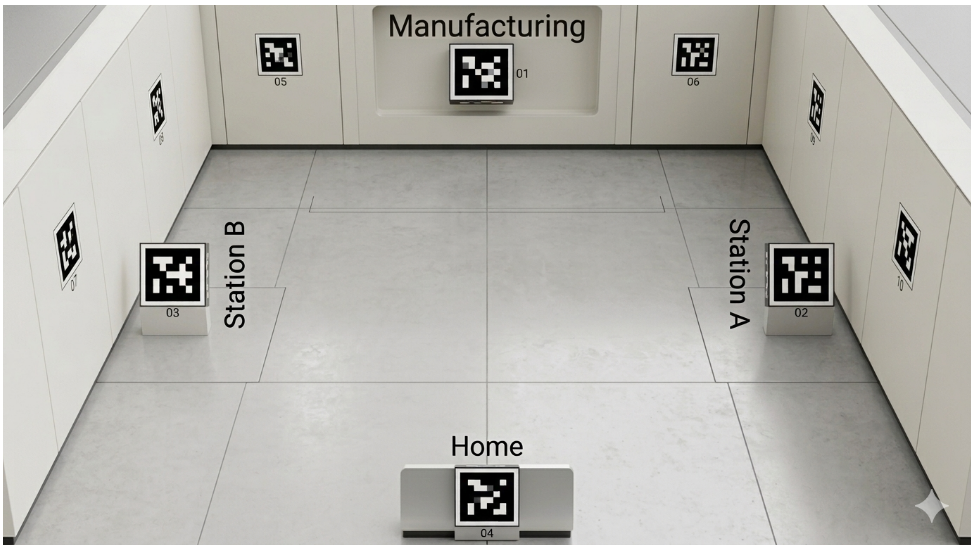


Figure 1: Visual exemple Warehouse environment used for the autonomous robot project.

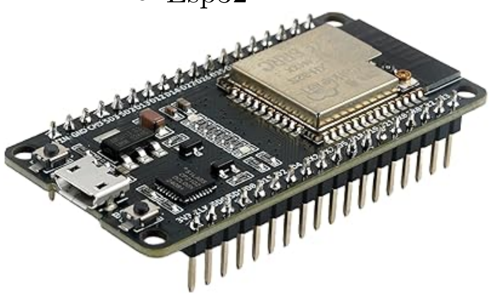
About Materials

- Raspberry pi

- Robot (to design ?)

- Gripper (with servo)

- Esp32



You will also need to consider the battery. You may potentially need to design additional mechanical parts, and you may also need to solder or glue components to obtain your final robot. The project is flexible; you can use other sensors if you want, or even different robots.